# WHY THE GOLDEN RATIO SELECTS THE PRIME THREE

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Abstract. For a positional base $b \geq 2$, a prime $p \nmid b$ with $b \equiv 1 \pmod{p}$, and a positive integer $m$ all of whose prime factors divide $b$, the repetend alignment in base $b$ satisfies

$$\alpha_b(pm) = \frac{2m-1}{pm-1}.$$

The formula's algebraic properties are therefore independent of the choice of base once the one-digit periodicity hypothesis $b \equiv 1 \pmod{p}$ is met.

We prove that $\alpha \geq 1/\varphi$ (the reciprocal of the golden ratio) if and only if $m \geq \varphi^2 = \varphi + 1$, provided $p = 3$. We then establish a stronger result. The self-referential condition $m^*(\tau) = 1/\tau^2$ yields a cubic equation that admits a root in $(0, 1)$ for every prime $p \geq 3$, but that factors over $\mathbb{Q}$ only for $p \in \{2, 3, 5\}$. The two non-trivial factorizations produce thresholds $1/\varphi$ and $1/\varphi^2$, both in the golden ratio's number field $\mathbb{Q}(\sqrt{5})$. The golden ratio's minimal polynomial $\tau^2 + \tau - 1$ divides the cubic if and only if $p = 3$, with coefficient 3 produced by polynomial division. For $p \geq 7$, the cubic is irreducible and the self-referential threshold is a cubic irrational outside any quadratic field.

The concrete setting is decimal arithmetic, where the alignment formula describes the coherence of fractional fields $\{k/n : 1 \leq k \leq n-1\}$. But the central results, the golden threshold, the self-referential characterization, and the algebraic selection of the prime 3, hold for any positional base $b$ in the one-digit periodic regime $b \equiv 1 \pmod{p}$; in particular, for $p = 3$ this includes every base with $b \equiv 1 \pmod{3}$ (including base 10).

## 1. Introduction

The decimal expansion of a rational number $k/n$ either terminates or eventually repeats. When it repeats, the shortest repeating block is called the *repetend*. The length of the repetend of $1/n$, for $\gcd(n, 10) = 1$, equals the multiplicative order $\mathrm{ord}_n(10)$, and the repetends of the family

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$\{k/n : \gcd(k,n) = 1\}$ are cyclic permutations of one another. These facts are classical [3].

Less studied is the following question. Given a fixed denominator $n$, consider the *full fractional field* $\mathcal{F}(n) = \{k/n : 1 \le k \le n-1\}$ and ask: what proportion of this field shares the repetend character of $1/n$? Some elements will have the same repeating digit or block; others will differ; still others will terminate altogether. The resulting proportion is a natural measure of *coherence*: the degree to which the arithmetic of $n$ organizes its fractional field into a single periodic pattern.

In this note we compute this measure exactly for a natural family of denominators, obtain a closed-form alignment formula, and discover that the golden ratio plays a structural role in the result. (This note predates the collision invariant series [5] and uses the earlier repetend/alignment notation rather than the digit-function conventions of that series.) The central theorems depend only on the alignment formula $\alpha = (2m-1)/(pm-1)$ and the algebra of the golden ratio, and are therefore independent of the choice of positional base. Decimal arithmetic provides the motivating instance, but the algebraic content is base-free.

## 2. Definitions

We work in base 10 throughout. For positive integers $k$ and $n$ with $1 \le k \le n-1$, the decimal expansion of $k/n$ takes the form

$$\frac{k}{n} = 0.a_1 a_2 \cdots a_s \overline{d_1 d_2 \cdots d_\ell},$$

where $a_1 \cdots a_s$ is the non-repeating prefix and $d_1 \cdots d_\ell$ is the repetend of length $\ell \ge 0$. When $\ell = 0$ the expansion terminates.

**Definition 1.** A positive integer is called $\{2,5\}$-*smooth* (or 10-*smooth*) if its only prime factors belong to $\{2,5\}$. Equivalently, $m$ is 10-smooth if and only if $1/m$ has a terminating decimal expansion. We write $\mathcal{S}_{10}$ for the set of all 10-smooth positive integers.

**Definition 2.** The *repetend alignment* of $n$ in base $b$ is the mean, over $k \in \{1, \ldots, n-1\}$, of the position-wise digit-match proportion between the repetend of $k/n$ and the repetend of $1/n$:

$$\alpha(n) = \frac{1}{n-1} \sum_{k=1}^{n-1} a(k/n,\, 1/n),$$

where $a(k/n,\, 1/n)$ is defined as follows: if $k/n$ terminates, $a = 1$; if $k/n$ has a repetend of the same length as $1/n$, $a$ is the fraction of positions at which the two repetends share the same digit; otherwise $a = 0$. In

the family $n = 3m$ studied here, the repetend of $1/n$ has length one, so $a$ reduces to exact single-digit matching.

*Remark* 3. For the family $n = 3m$ with $m \in \mathcal{S}_{10}$, the repetend of $1/n$ has length one. In this case, the position-wise definition reduces to exact digit matching: $a = 1$ if the repeating digit agrees, $a = 0$ otherwise. The general position-wise definition is used throughout the series to handle longer repetends at other primes.

## 3. The Three Classes

The following lemma establishes the arithmetic that underlies the alignment count.

**Lemma 4.** *Let $n = 3m$ with $m \in \mathcal{S}_{10}$ and $m \geq 1$. For each $k \in \{1, \ldots, n-1\}$, exactly one of the following holds:*

(A) $k \equiv 0 \pmod 3$. *Then $k/n = (k/3)/m$, and since $m \in \mathcal{S}_{10}$ the expansion terminates.*
(B) $k \equiv 1 \pmod 3$. *Then $k/n$ has repetend of length one, equal to the repeating digit of $1/n$.*
(C) $k \equiv 2 \pmod 3$. *Then $k/n$ has repetend of length one, equal to the nines complement of the repeating digit of $1/n$.*

*Proof.* Write $k = 3q + r$ with $r \in \{0, 1, 2\}$. Then

$$\frac{k}{3m} = \frac{q}{m} + \frac{r}{3m}.$$

Since $m \in \mathcal{S}_{10}$, the term $q/m$ terminates. Adding a terminating decimal does not change the repetend.

*Case* $r = 0$. Then $k/(3m) = q/m$, which terminates.

*Case* $r = 1$. The repetend of $k/(3m)$ equals the repetend of $1/(3m)$, which is the repeating digit of $1/n$. (Since $\mathrm{ord}_3(10) = 1$, the repetend has length one.)

*Case* $r = 2$. The repetend of $k/(3m)$ equals the repetend of $2/(3m)$. Since $1/(3m)+2/(3m) = 1/m$ terminates, the repeating digits of $1/(3m)$ and $2/(3m)$ are nines complements: they sum to 9. □

*Remark* 5. The nines complement relationship $d^* + (9 - d^*) = 9$ between classes (B) and (C) reflects the identity $1/3 + 2/3 = 1$ and its decimal manifestation $0.\overline{3} + 0.\overline{6} = 0.\overline{9} = 1$. This complement symmetry is a special case of the universal identity $k/n + (n-k)/n = 1$, which pairs every element of $\mathcal{F}(n)$ with its complement. In base 10 the digit pairs $\{0, 9\}$, $\{1, 8\}$, $\{2, 7\}$, $\{3, 6\}$, $\{4, 5\}$ each sum to 9, and the alignment

classes (B) and (C) correspond precisely to the complement pair $\{3, 6\}$ associated with the prime 3.

## 4. The Alignment Formula

**Theorem 6.** *Let $n = 3m$ with $m \in \mathcal{S}_{10}$ and $m \geq 1$. Then*

$$\alpha(n) = \frac{2m-1}{3m-1}.$$

*Proof.* By Lemma 4 the aligned fractions comprise classes (A) and (B). The counts are as follows.

*Class (A):* The multiples of 3 in $\{1, \ldots, 3m-1\}$ are $3, 6, \ldots, 3(m-1)$, giving $m-1$ elements.

*Class (B):* The integers $k \equiv 1 \pmod 3$ in $\{1, \ldots, 3m-1\}$ are $1, 4, 7, \ldots, 3m-2$, giving $m$ elements.

*Class (C):* The remaining $m$ elements are not aligned.

The total count of aligned elements is $(m-1) + m = 2m - 1$ out of $3m - 1$, and the formula follows. □

**Corollary 7.** *As $m \to \infty$ through $\mathcal{S}_{10}$,*

$$\alpha(3m) \longrightarrow \frac{2}{3}.$$

The convergence is monotone from below. The first few values are displayed in Table 1.

| $m$ | $n = 3m$ | $\alpha(n)$ | Decimal |
|---|---|---|---|
| 1 | 3 | 1/2 | 0.500000 |
| 2 | 6 | 3/5 | 0.600000 |
| 4 | 12 | 7/11 | 0.636364 |
| 5 | 15 | 9/14 | 0.642857 |
| 8 | 24 | 15/23 | 0.652174 |
| 10 | 30 | 19/29 | 0.655172 |
| 16 | 48 | 31/47 | 0.659574 |
| 32 | 96 | 63/95 | 0.663158 |
| 100 | 300 | 199/299 | 0.665552 |
| 256 | 768 | 511/767 | 0.666232 |

Table 1. Repetend alignment $\alpha(3m)$ for selected 10-smooth values of $m$.

## 5. The Golden Threshold

We now ask: for which 10-smooth integers $m$ does the alignment exceed the reciprocal of the golden ratio $\varphi = (1+\sqrt{5})/2$?

**Theorem 8.** *Let $m \geq 1$ be a positive integer. Then*

$$\frac{2m-1}{3m-1} \geq \frac{1}{\varphi} \qquad \Longleftrightarrow \qquad m \geq \varphi^2.$$

*Since $\varphi^2 = \varphi + 1 = (3+\sqrt{5})/2 \approx 2.618$, the integer threshold is $m \geq 3$.*

*Proof.* The inequality $(2m-1)/(3m-1) \geq 1/\varphi$ is equivalent to

$$\varphi(2m-1) \geq 3m-1.$$

Expanding and collecting terms:

$$m(2\varphi - 3) \geq \varphi - 1.$$

Since $\varphi - 1 = 1/\varphi$ and $\varphi^2 = \varphi + 1$, we compute

$$2\varphi - 3 = 2\varphi - 3 = \frac{2(1+\sqrt{5})-6}{2} = \frac{2\sqrt{5}-4}{2} = \sqrt{5}-2.$$

This is positive (since $\sqrt{5} > 2$), so we may divide:

$$m \geq \frac{1/\varphi}{\sqrt{5}-2} = \frac{1}{\varphi(\sqrt{5}-2)}.$$

Now $\varphi(\sqrt{5}-2) = \varphi\sqrt{5} - 2\varphi$. Using $\varphi = (1+\sqrt{5})/2$:

$$\varphi\sqrt{5} = \frac{(1+\sqrt{5})\sqrt{5}}{2} = \frac{\sqrt{5}+5}{2}, \qquad 2\varphi = 1+\sqrt{5},$$

so

$$\varphi(\sqrt{5}-2) = \frac{\sqrt{5}+5}{2} - (1+\sqrt{5}) = \frac{\sqrt{5}+5-2-2\sqrt{5}}{2} = \frac{3-\sqrt{5}}{2}.$$

Therefore

$$m \geq \frac{2}{3-\sqrt{5}} = \frac{2(3+\sqrt{5})}{(3-\sqrt{5})(3+\sqrt{5})} = \frac{2(3+\sqrt{5})}{4} = \frac{3+\sqrt{5}}{2} = \varphi^2. \quad \square$$

The identity $\varphi^2 = \varphi + 1$ is the defining property of the golden ratio [4]. Its appearance here is not assumed but *derived*: it emerges from the interaction between the golden threshold $1/\varphi$ and the ternary structure of residue classes modulo 3.

**Corollary 9.** *Among denominators of the form $n = 3m$ with $m \in \mathcal{S}_{10}$, the repetend alignment $\alpha(n)$ exceeds $1/\varphi$ if and only if $m \geq 4$ (the smallest 10-smooth integer at least 3).*

*Proof.* By Theorem 8 the integer threshold is $m \geq 3$. The value $m = 3$ is not 10-smooth (since $3 \notin \{2, 5\}$), so the smallest admissible value is $m = 4 = 2^2$. □

The first admissible denominator is therefore $n = 12$, and $\alpha(12) = 7/11 \approx 0.6364$, above $1/\varphi \approx 0.6180$.

**The qualifying families.** Two families of integers are proved to satisfy $\alpha(n) \geq 1/\varphi$:

**Tier 1.** $n \in \mathcal{S}_{10}$. Every fraction $k/n$ terminates, so $\alpha(n) = 1$.

**Tier 2.** $n = 3m$ with $m \in \mathcal{S}_{10}$ and $m \geq 4$. Alignment satisfies $1/\varphi \leq \alpha(n) < 1$ by Theorem 6 and Theorem 8.

Exhaustive computation up to $n \leq 909$ finds no integer outside these two families satisfying $\alpha(n) \geq 1/\varphi$. The following conjecture asserts that the two families above form the complete classification.

**Conjecture 10.** For every positive integer $n$, the repetend alignment satisfies $\alpha(n) \geq 1/\varphi$ if and only if $n \in \mathcal{S}_{10}$ or $n = 3m$ with $m \in \mathcal{S}_{10}$ and $m \geq 4$.

*Note added April 2026: Conjecture 10 was subsequently proved by the author as the three-tier theorem ("The Three-Tier Theorem," research note, 2022). The expected mechanism, sketched below, is the proof that paper uses.*

The expected mechanism is the nines complement: non-terminating fractions $k/n$ and $(n-k)/n$ have complementary digits at every position, so their alignment contributions cannot overlap, which bounds $\alpha(n) \leq (t+1)/(2t)$ for rough part $t$. This bound falls below $1/\varphi$ for $t \geq 7$.

## 6. Why the Golden Ratio, and Why the Prime Three

The result of Theorem 8 invites the question: is there something special about the threshold $1/\varphi$, or would any threshold between 3/5 and 7/11 produce a similar classification?

A threshold $\tau$ separates the family $\{3m : m \in \mathcal{S}_{10}\}$ into aligned and non-aligned members if and only if it lies in the gap between $\alpha(6) = 3/5$ and $\alpha(12) = 7/11$. Since $3/5 < 1/\varphi < 7/11$, the golden ratio does lie in this gap. In this narrow sense, any $\tau \in (3/5, 7/11)$ would work.

What distinguishes $1/\varphi$ is the algebraic structure of the threshold equation. The next theorem formalizes the prime-uniform extension that underlies the discussion.

**Theorem 11** (General prime threshold)**.** *Let $b \geq 2$, let $p \nmid b$ be a prime with $b \equiv 1 \pmod{p}$, and let $m$ be a positive integer all of whose prime*

*factors divide $b$. Let $\alpha_b$ denote repetend alignment computed in base $b$. Then*

$$\alpha_b(pm) = \frac{2m-1}{pm-1}.$$

*Consequently,*

$$\alpha_b(pm) \geq \frac{1}{\varphi}$$

*holds for some positive integer $m$ if and only if $p \in \{2,3\}$. Among odd primes, this occurs if and only if $p = 3$, in which case the threshold is*

$$m \geq \varphi^2 = \varphi + 1.$$

*For every prime $p \geq 5$, no positive integer $m$ satisfies $\alpha_b(pm) \geq 1/\varphi$.*

*Proof.* Choose $s \geq 0$ with $m \mid b^s$, and write $u = b^s/m$. Then

$$\frac{k}{pm} = \frac{ku}{p\,b^s}.$$

Thus the base-$b$ expansion of $k/(pm)$ is obtained from that of $ku/p$ by shifting the radix point $s$ places. In particular, the repetend of $k/(pm)$ is determined by the residue of $ku \pmod p$.

Since $b \equiv 1 \pmod p$, one has $\mathrm{ord}_p(b) = 1$, so every nonzero residue class modulo $p$ yields a one-digit repetend. Also $p \nmid u$, so multiplication by $u$ permutes the nonzero residue classes modulo $p$.

Write $k = pq + r$ with $r \in \{0, 1, \ldots, p-1\}$. If $r = 0$, then $k/(pm) = q/m$ terminates because all prime factors of $m$ divide $b$. If $r \neq 0$, then

$$ku \equiv ru \pmod p,$$

and the map $r \mapsto ru \pmod p$ is a bijection on $\{1, \ldots, p-1\}$. Therefore the $p-1$ nonzero residue classes modulo $p$ produce $p-1$ distinct one-digit repetends. The class $r = 1$ produces the same repetend as $1/(pm)$, while the classes $r = 2, \ldots, p-1$ do not.

So among the integers $k \in \{1, \ldots, pm-1\}$, exactly the following are aligned with $1/(pm)$:

(1) the $m-1$ multiples of $p$, which terminate;
(2) the $m$ integers congruent to $1 \pmod p$.

Hence the aligned count is $(m-1) + m = 2m-1$ out of $pm-1$ total, giving

$$\alpha_b(pm) = \frac{2m-1}{pm-1}.$$

For the threshold, the inequality

$$\frac{2m-1}{pm-1} \geq \frac{1}{\varphi}$$

is equivalent to

$$m(2\varphi - p) \geq \varphi - 1.$$

If $p < 2\varphi \approx 3.236$, then $p \in \{2, 3\}$. For $p = 2$ the threshold is $m \geq (\varphi - 1)/(2\varphi - 2) = 1/2$, so every positive $m$ works. For $p = 3$, the threshold is

$$m \geq \frac{\varphi - 1}{2\varphi - 3} = \varphi^2.$$

If $p \geq 5$, then $2\varphi - p < 0$, so dividing reverses the inequality and yields

$$m \leq \frac{\varphi - 1}{2\varphi - p} < 0,$$

which has no positive integer solutions. □

The prime $p = 3$ is therefore the *unique* odd prime for which the golden threshold produces a non-trivial classification, and the critical resolution $\varphi^2 = \varphi + 1$ is the signature of this uniqueness.

## 7. The Self-Referential Cubic and the Golden Field

The preceding section establishes that $1/\varphi$ is the unique golden threshold selecting exactly one non-trivial prime. We now show that the golden ratio has a deeper algebraic role in the self-referential structure of the alignment formula.

**Definition 12.** For a threshold $\tau \in (0, 1)$ and a prime $p$, the *critical resolution* is the value $m^*(\tau, p)$ at which the alignment $(2m-1)/(pm-1)$ equals $\tau$:

$$m^*(\tau, p) = \frac{1 - \tau}{2 - p\tau}.$$

We say the classification is *self-referential* at $(\tau, p)$ if the critical resolution equals the reciprocal square of the threshold:

$$m^*(\tau, p) = \frac{1}{\tau^2}.$$

The self-referential condition requires $\tau$ to satisfy the cubic

$$\tau^3 - \tau^2 - p\tau + 2 = 0. \tag{1}$$

Since $f(\tau) = \tau^3 - \tau^2 - p\tau + 2$ satisfies $f(0) = 2 > 0$ and $f(1) = 2 - p < 0$ for every prime $p \geq 3$, the intermediate value theorem guarantees a root in $(0, 1)$ for every such $p$. The self-referential condition therefore admits a solution for every prime $p \geq 3$.

What distinguishes $p = 3$ is not the existence of a solution but its algebraic nature. The following theorem shows that the self-referential cubic factors over $\mathbb{Q}$ for exactly three primes, and the two non-trivial cases both produce thresholds in the golden ratio's number field.

**Theorem 13.** *The self-referential cubic* (1) *factors over* $\mathbb{Q}$ *if and only if* $p \in \{2, 3, 5\}$. *The factorizations are:*

$$p = 2: \quad (\tau - 1)(\tau^2 - 2) = 0, \tag{2}$$

$$p = 3: \quad (\tau - 2)(\tau^2 + \tau - 1) = 0, \tag{3}$$

$$p = 5: \quad (\tau + 2)(\tau^2 - 3\tau + 1) = 0. \tag{4}$$

*For* $p \geq 7$, *the cubic is irreducible over* $\mathbb{Q}$.

*Proof.* By the rational root theorem, the only possible rational roots of $\tau^3 - \tau^2 - p\tau + 2$ are $\pm 1$ and $\pm 2$. Evaluating:

$$f(1) = 2 - p, \quad f(-1) = p, \quad f(2) = 6 - 2p, \quad f(-2) = 2p - 10.$$

These vanish at $p = 2, 3, 5$ respectively ($f(-1) = p$ is never zero for a prime). For $p \geq 7$, no rational root exists and the cubic is irreducible. □

The case $p = 2$ yields $\tau = 1$ (boundary of the interval) and $\tau = \sqrt{2}$ (outside $(0, 1)$), so it produces no self-referential threshold. The two non-trivial factorizations are $p = 3$ and $p = 5$.

**Theorem 14.** *The self-referential thresholds for* $p = 3$ *and* $p = 5$ *are*

$$\tau(3) = \frac{\sqrt{5} - 1}{2} = \frac{1}{\varphi}, \qquad \tau(5) = \frac{3 - \sqrt{5}}{2} = \frac{1}{\varphi^2}.$$

*Both lie in the quadratic number field* $\mathbb{Q}(\sqrt{5})$, *the field generated by the golden ratio. They are consecutive powers of* $1/\varphi$.

*For* $p \geq 7$, *the self-referential threshold is a cubic irrational and does not lie in any quadratic number field.*

*Proof.* The quadratic factors in (3) and (4) have discriminants $1 + 4 = 5$ and $9 - 4 = 5$ respectively, so both roots lie in $\mathbb{Q}(\sqrt{5})$. For $p = 3$, the root in $(0, 1)$ is $(\sqrt{5} - 1)/2 = 1/\varphi$. For $p = 5$, it is $(3 - \sqrt{5})/2$. Since $(3 - \sqrt{5})/2 = ((\sqrt{5} - 1)/2)^2$ (verified by expanding), this equals $1/\varphi^2$.

For $p \geq 7$, the cubic is irreducible over $\mathbb{Q}$ by Theorem 13, so its roots generate a cubic extension of $\mathbb{Q}$ and cannot lie in any quadratic field. □

**Corollary 15.** *The golden ratio's minimal polynomial* $\tau^2 + \tau - 1$ *divides the self-referential cubic if and only if* $p = 3$. *The resulting fourth-power identity is*

$$\tau^4 = 2 - 3\tau,$$

*where the coefficient* 3 *is the unique prime annihilating the remainder of the polynomial division.*

*Proof.* Dividing $\tau^3 - \tau^2 - p\tau + 2$ by $\tau^2 + \tau - 1$ yields quotient $\tau - 2$ and remainder $(3-p)\tau$. The remainder vanishes if and only if $p = 3$. The fourth-power identity follows from $\tau^2 = 1 - \tau$. □

The self-referential structure therefore reveals a hierarchy among primes:

(1) $p = 3$: threshold $1/\varphi$, governed by the golden ratio's own minimal polynomial. The alignment limit $2/3$ exceeds this threshold, producing the classification of Theorem 8.
(2) $p = 5$: threshold $1/\varphi^2$, governed by the companion quadratic $\tau^2 - 3\tau + 1$. The alignment limit $2/5$ exceeds this threshold, but falls below $1/\varphi$, so no golden classification arises.
(3) $p \geq 7$: thresholds are cubic irrationals outside $\mathbb{Q}(\sqrt{5})$. No connection to the golden ratio persists.

The golden ratio's number field $\mathbb{Q}(\sqrt{5})$ controls exactly two primes in the self-referential hierarchy.

## 8. Remarks

**Base independence.** The alignment formula $\alpha = (2m-1)/(pm-1)$ and the self-referential characterization of Theorem 14 are purely algebraic and make no reference to a positional base. The base enters only in determining which prime $p$ governs the periodicity of $1/n$. In any base $b$ satisfying $b \equiv 1 \pmod 3$ (such as 4, 7, 10, 13, 16, or 19), the prime 3 plays this role and Theorem 11 applies with "10-smooth" replaced by "$b$-smooth." In bases where 3 divides $b$ (such as 6, 12, or 30), a different prime governs periodicity. Since $p = 3$ is the unique prime selected by the golden ratio's algebra, the theorem does not apply to such bases. The result is therefore base-free in its algebraic content and base-dependent only in whether its hypotheses are met.

**Terminating fractions as aligned.** The decision to count terminating fractions as aligned is not mathematically forced. If terminating fractions are instead excluded, the alignment among purely periodic fractions is exactly $1/2$ for every $m$, and the golden ratio plays no role. The classification theorem therefore reflects a specific but natural convention: that the absence of a repeating obstruction constitutes compatibility.

**Connection to Midy's theorem.** Midy's theorem [1, 2] and its generalizations study digit-sum relations within a single repetend, typically for prime denominators. The present result addresses a complementary question: not the internal structure of one repetend, but the collective

organization of an entire fractional field. The nines complement relationship between classes (B) and (C) is, however, a manifestation of the same arithmetic that underlies Midy's observation.

**Computational origin.** This result was discovered through computational exploration of the fractional fields $\mathcal{F}(n)$ for $n \leq 909$, using software that computes structural invariants of decimal expansions [5]. The empirical observation that a $1/\varphi$ threshold cleanly separates a family of denominators was observed computationally; the algebraic proof came later.

## References

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